Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when assumptions apply.
3. Explain energy and model limits clearly.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to explain your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

conserved: unchanged in total under the stated conditions

01 Bring back an earlier idea

In the same straight-line deflection model, keep sideways Δv fixed but make lead time 3 times as long. **Compare** shift and name a limitation.

Word help: deflection = a change in the direction or path of motion

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$. Use momentum conservation only when external impulse is negligible. Label before/after diagrams and directions.

Worked reasoning: A 2 kg cart moving at 3 m/s stops in 0.5 s. Momentum change is -6 kg·m/s; average force is -12 N. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

A 1 kg cart at 4 m/s right sticks to a 1 kg cart at rest. **Draw** before/after and explain which quantity is conserved.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart slows from 6 m/s right to rest in 5 s. **Find** momentum change size and average stopping force size.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A report uses equal total momentum before/after to claim a collision is elastic. What additional calculation is required?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg cart travels left at 6 m/s.
Find momentum and draw its direction.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A fixed impulse gives a target .04 m/s sideways velocity change.
Find shift after 500 s and state a model limitation.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2.
Design two stopping processes for this cart that give the same impulse but different average forces.
Explain the time-force tradeoff.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3